CLINICAL · EAR

Tuberculous and syphilitic otitis media

Tuberculous otitis media hides in plain sight as an ordinary chronic ear that simply will not get better. The clues are a painless, thin discharge, several holes in a thickened drum, pale granulation, and a hearing loss worse than the symptoms suggest. Send the tissue to the laboratory, because the treatment is medical, not surgical.

AT A GLANCE

- Painless discharge, multiple perforations and pale granulation.
- Hearing is worse than symptoms suggest.
- Confirm by PCR, histology and culture.
- Antituberculous drugs first; repair later.

1 CHAPTER ONE Overview

The chronic ear that refuses to heal.

1.1 Definition

Infection of the middle ear cleft by *Mycobacterium tuberculosis*. Syphilitic otitis media, much rarer, is included here because it shares the picture of bone destruction and a disproportionate hearing loss.

1.2 Epidemiology

Uncommon in low-burden countries, but important where tuberculosis is endemic, and rising with drug-resistant strains and immunosuppression, particularly HIV.

1.3 Routes of infection

- Through the Eustachian tube, from active pulmonary disease, which most patients have.
- Through the blood, from a distant focus.

- In infants, through the tube from infected milk.

2 CHAPTER TWO Clinical features

Painless, thin, and resistant.

2.1 Symptoms

- **Discharge:** painless, thin and cloudy, persisting despite treatment.
- **Hearing loss:** out of proportion to the symptoms, often mixed, from early ossicular and cochlear involvement.

2.2 Signs

- A thickened drum early on, then *multiple perforations* that merge into one large defect.
- Pale, thick mucosa and abundant pale granulation.
- Bone sequestra in the mastoid.
- Non-tender enlarged cervical and preauricular nodes.
- Facial palsy in about 10%, sometimes the presenting feature in a child.

2.3 Complications

Facial palsy, sequestrum, postauricular fistula, Bezold abscess and labyrinthitis.

> **SUSPECT IT WHEN**
>
> A chronic ear fails ordinary treatment, a mastoidectomy fails, granulation is pale, or facial palsy appears early in a painless ear.

3 CHAPTER THREE Investigations

Prove it.

3.1 From the ear

- **PCR of the discharge or tissue:** gives an answer within days.
- **Histology:** caseating granulomas with Langhans giant cells.
- **Acid-fast smear and culture:** a negative smear does not exclude it.

3.2 General

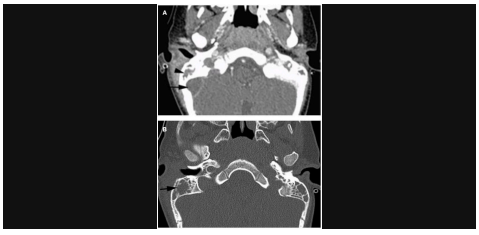
Chest imaging, tuberculin skin test or interferon-gamma release assay, and HIV testing.

3.3 CT

Often looks like ordinary chronic otitis media, with more bone erosion and sequestra.

IMAGING IN THE APP 1 case

Otomastoiditis on CT.



Otomastoiditis and its complications

Infection & inflammation · 9 images

3.4 Differential diagnosis

Ordinary chronic suppurative otitis media, carcinoma of the middle ear, granulomatosis with polyangiitis, and fungal infection.

4 CHAPTER FOUR Management

Drugs first, repair later.

4.1 Antituberculous chemotherapy

Standard multidrug therapy with isoniazid, rifampicin, pyrazinamide and ethambutol, guided by sensitivities, for at least six months.

4.2 Surgery

Limited to biopsy, removal of sequestra and drainage of abscesses or complications. Reconstruction is delayed until treatment is complete, because healing is slow and fistulae are common during active disease; afterwards, repair succeeds in most.

OPERATIVE GUIDES IN THEATRE 1 operation

Myringoplasty and Tympanoplasty
Late repair

4.3 Facial palsy

Usually recovers well with medical treatment.

5 CHAPTER FIVE Syphilitic otitis

Rare, destructive.

Congenital or acquired syphilis can cause osteitis of the temporal bone with necrosis, and labyrinthitis with profound, often fluctuating sensorineural loss and vertigo. Diagnose with treponemal tests (FTA-ABS) and treat with penicillin and steroids.

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The features:** painless discharge, multiple perforations, pale granulation, early facial palsy, disproportionate hearing loss.
- **Diagnosis** by PCR, histology and culture.
- **Treatment is medical;** surgery for complications only.

6.2 Common pitfalls

COMMON PITFALLS

- Operating on a chronic ear without sending the tissue for tuberculosis.
- Being reassured by a negative acid-fast smear.
- Reconstructing during active treatment.

Written for this app from Oxford Textbook of Otolaryngology ch. 8, 24, Oxford Specialist Handbook ch. 16, Dhingra ch. 11 and K. J. Lee ch. 22. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

MIDDLE EAR & MASTOID · INFECTION & INFLAMMATION

Otomastoiditis and its complications

IMAGING FINDINGS

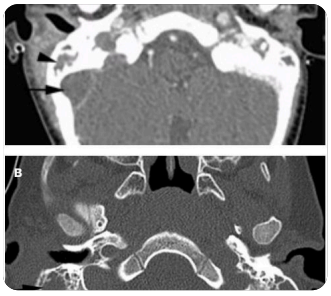
- Opacified middle ear and mastoid
- Coalescence: loss of the mastoid septa and cortical erosion
- Complications: subperiosteal abscess, sigmoid sinus thrombosis, epidural or brain abscess, labyrinthitis

TOP DIFFERENTIALS

- Uncomplicated effusion (septa intact)
- Cholesteatoma
- Langerhans cell histiocytosis or rhabdomyosarcoma in a child

CHECKLIST

- Contrast CT or MRI with venography when complications are suspected
- Check the sigmoid sinus, tegmen, labyrinth and petrous apex



Post-contrast CT shows (a) right coalescent mastoiditis (black arrowhead) and ipsilateral sigmoid sinus thrombosis (black arrow) – bot…



Axial CECT in a patient with postauricular tender mass, headache, and fever reveals postauricular abscess and coalescent…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1042.